

PAH-OMC-019: Minimal form closure and radial nullspace

TECT verification-first repository
claim C6-SPACETIME-SIGNATURE

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Purpose and scope

This checkpoint concerns the exact R-511 pre-form on the invariant R-510 state Hilbert space. It adds no model, rate, projection or time rescaling. The PAH parameters remain $K = 2$, $M_s = 1$, $\epsilon = 1/2$, $\beta = \nu = 1$, $m^2 = \theta = 0$, with all other OMC-016 displayed couplings one. The anchored two-row strip has $2(n+2)$ vertices and its unsplit frontier square remains. The full labelled all- Q law has weights Z_Q/Z . The state uses $h_j = 2^{-j}$, $R_j = 2^j$, $M_j = 2^{2j}$, first j at fixed n , then the specified anchored n exhaustion. Closure below is on that fixed state; it is not another regulator limit or a physical continuum construction.

The preregistration preceded checks:

```
strategy/pa-hyp/PAH-OMC-019-closure-prereg-v1.json  
4337e21a140206956ea52ee10e20d04358af868969e3edb9f055e5ea64b055b7
```

The full proof, hypothesis crosswalk and falsifiers are in the companion closure certificate. This is its one gate-level synthesis, not a replacement for the inherited R-509–R-511 measure and local-generator arguments.

Exact domain and result

Let H be the real invariant R-510 cylinder-state Hilbert space modulo zero L^2 norm. Let D be exactly the R-511 bounded, gauge-invariant, globally amplitude- ℓ^1 -Lipschitz finite-prefix cylinder domain. R-511 gives

$$A : D \longrightarrow H, \quad \|Ag\|_H \leq 2D_gM_g, \quad E(f, g) = -\langle f, Ag \rangle_H = \frac{1}{2} \sum_r \nu(c_r \Delta_r f \Delta_r g).$$

Here A is the derived original PH/LK/AP sum, with the same partial domains, rates and label multiplicities. The sum is finite on each fixed local pair. The form is nonnegative and symmetric, D is dense, and amplitude-only elements of D have zero A . None of these are inferred merely from finite fixture agreement. Write $S = -A$, D_{rad} for the amplitude-only subspace and $H_{\text{rad}} = \overline{D_{\text{rad}}}^{\|\cdot\|_H}$.

Scoped theorem. The exact form E is closable. Its minimal closed nonnegative extension $\overline{E}$ has dense domain and satisfies

$$H_{\text{rad}} \subset \text{Dom}(\overline{E}), \quad \overline{E}(h, h) = \overline{E}(h, u) = 0 \quad (h \in H_{\text{rad}}, u \in \text{Dom}(\overline{E})).$$

The original aperture witness retains strictly positive energy. This does not classify the complete kernel and does not construct a limit path law.

Applicability and quotient check

The standard nonnegative symmetric-operator form construction is used; it is not claimed as a new theorem. The primary reference is G. Teschl, *Mathematical Methods in Quantum Mechanics*, second edition, Section 2.3, printed pp. 77–81, equations (2.47)–(2.49), Theorem 2.13; author-hosted text consulted 2026-09-07. Theorem 2.14 assumes closedness and is not used circularly. Density, symmetry, nonnegativity and $Sg \in H$ match. Self-adjointness, $S(D) \subset D$, bounded global exit rate and a uniform operator norm are not required. The quantum interpretation of the reference does not apply.

First let $x \in D$ be zero in H . For every $y \in D$,

$$\langle y, Sx \rangle = E(y, x) = E(x, y) = \langle x, Sy \rangle = 0.$$

Density implies $Sx = 0$. Thus both S and E descend to the state quotient; the operator theorem is not applied to an ill-defined representative map.

Two closability proofs

Positivity of $E(x + ty, x + ty)$ for all real t gives form Cauchy–Schwarz. Let K be the Hilbert completion of $D/\ker E$ in the energy norm and let $j : D \rightarrow K$ be its dense natural map. The exact pairing is

$$\langle jf, jg \rangle_K = E(f, g) = \langle f, Sg \rangle_H. \quad (1)$$

For any $f_n \rightarrow 0$ in H which is energy-Cauchy, completeness gives $jf_n \rightarrow v$ in K . For each fixed $g \in D$, (1) gives $\langle v, jg \rangle = \lim_n \langle f_n, Sg \rangle = 0$. Density implies $v = 0$, hence $E(f_n, f_n) \rightarrow 0$. Supports may grow arbitrarily: neither a common support nor a bound on $\|Sf_n\|$ was used.

Independently, the form seminorms of an energy-Cauchy sequence are bounded by M . For $\varepsilon > 0$ choose N with $\sqrt{E(f_n - f_m)} < \varepsilon$ for $m, n \geq N$, and fix $m \geq N$. Then

$$E(f_n, f_n) \leq M\varepsilon + |E(f_n, f_m)| = M\varepsilon + |\langle f_n, Sf_m \rangle|.$$

The last term tends to zero against the fixed $Sf_m \in H$. Arbitrary ε proves the same criterion. Taking $m = n$ would be an invalid shortcut; the hostile audit explicitly tests that error.

Minimal closure and the entire radial subspace

Complete D in $\|f\|_1^2 = \|f\|_H^2 + E(f, f)$ to V . Its contractive map $i : V \rightarrow H$ is injective: a completion vector over zero is represented by an H -null energy-Cauchy sequence, so closability makes its form norm tend to zero. On $\overline{D} = i(V)$ define

$$\overline{E}(iu, iv) = \langle u, v \rangle_V - \langle iu, iv \rangle_H.$$

It is nonnegative, agrees with E , and its form norm is the complete V norm. Any other closed nonnegative extension agreeing on D contains the limits of all these form-Cauchy sequences and agrees on $\overline{D}$. This proves minimality, not equality with a maximal jump-form domain or uniqueness of all self-adjoint extensions.

For every $h \in H_{\text{rad}}$ choose $a_n \in D_{\text{rad}}$ with $a_n \rightarrow h$ in H . Every difference $a_n - a_m$ has zero energy, so it is form-Cauchy. Consequently $h \in \overline{D}$ and $\overline{E}(h, h) = 0$. Form Cauchy–Schwarz gives zero cross-energy against all of $\overline{D}$. The form norm on H_{rad} equals its H norm. Finite-prefix bounded Lipschitz functions are dense in the spatial amplitude L^2 space by compact approximation and the cylinder monotone-class argument. Thus this is the whole amplitude-only L^2 subspace, not a finite list of witnesses. No conditional averaging of the generator is introduced.

Since the source aperture cylinder s_v belongs to the original domain, $\overline{E}(s_v, s_v) = E(s_v, s_v) > 0$. Positive radial variance and zero radial energy coexist with nonzero label activity at the unchanged time scale.

Devil’s-advocate

1. Positivity alone implies closability: UPHELD against that shortcut. For $q(f) = f(0)^2$ on $C[0, 1] \subset L^2[0, 1]$, the tents $f_n = \max(1 - nx, 0)$ have $\|f_n\|_2^2 = 1/(3n)$, $q(f_n) = 1$ and $q(f_n - f_m) = 0$. This is a control, not a PAH counterexample. A fixed L^2 representer is precisely missing.
2. $\|f_n\| \rightarrow 0$ controls the diagonal pairing: UPHELD objection. $Se_n = n^2 e_n$, $f_n = e_n/n$ gives unit energy but is not energy-Cauchy. The proof pairs only with fixed test vectors.
3. Quotient or domain mismatch: DISMISSED under the pinned R-511 inputs by the explicit density and symmetry argument; no graph-core assumption.
4. A different closed extension might add vectors: VALID with mitigation. Only the minimal completion is used; maximal-domain equality remains open.
5. Radial retention implies a complete kernel classification: UPHELD objection. Additional nonradial null modes are not excluded.

Verification boundary and next question

Nine Lean declarations check the universal no-ghost/Cauchy-sequence argument, energy convergence, quotient representer, completion injectivity, radial closed-graph passage and zero-energy cross consequences. Model measure construction, analytic completion applicability and amplitude density are not fully formalized. Non-importing symbolic checks audit the direct epsilon proof; exact hostile controls attack the missing assumptions. These are internal cross-checks, not an external signed referee report.

The next single question is whether the original finite stationary semigroups converge, with the fixed comparison maps and j -before- n order, to the semigroup selected by this minimal closed form. That requires a separately activated contract, topology and bounded review budget. No temporal construction or convergence is claimed here. Re-review after this one attempt, or upon any source, state, density or representer defect.

Result footer

Result ID: R-512; contract PAH-OMC-019
Claim ID: C6-SPACETIME-SIGNATURE
Precise statement: Closable exact pre-form; minimal closure retains all H_{rad} at zero energy, with positive original aperture energy.
Scope: R-510 H and R-511 D, A, E; unchanged j -before- n path.
Dependencies: Exact inherited R-510 state and R-511 local pre-form.
Evidence grade: ANALYTIC, EXACT, EXECUTED; abstract Lean consequences.
Reproduction command: `python -X utf8 verification/scripts/pah_omc019_verify.py --check`
Expected output: PAH-OMC-019 INTEGRATED: PASS; Lean PASS.
Falsification gate: State, density, symmetry or representer defect; nonzero completion ghost or failure of radial closure passage.
Tier before / after: T1 / T1; auxiliary_support; T-054 unchanged.
No-overclaim statement: No full kernel or temporal limit theorem; no physical Pre-A, spacetime, QFT, gravity, continuum or TOE result.
Next required action: Separately contract finite-semigroup selection and convergence; no next temporal attempt activated here.